

Winter School in Mathematical Physics

Les Diablerets

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From matrix models to LG gravity:
the simplest gauge/string duality

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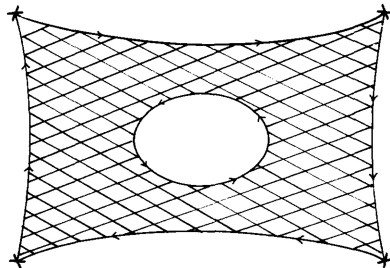
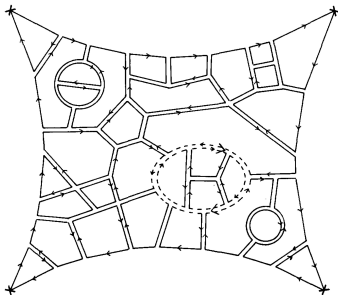
ETH Zürich

What is the worldsheet theory of a matrix model?

Both **matrix models** and **string theories** have a genus expansion.

Are they related?

If so, what is the closed string dual to a matrix model?



From: "A planar diagram theory for strong interactions"
G. 't Hooft, 1974

Matrix models

Partition function:

$$Z_N = \int dM e^{-N \text{Tr} V(M)}$$

- M : $N \times N$ Hermitian matrix
- $V(M)$: potential

Operators: single-trace operators $\text{Tr} M^k$

Correlators:

$$\left\langle \prod_{i=1}^n \text{Tr} M^{k_i} \right\rangle = \frac{1}{Z_N} \int dM e^{-N \text{Tr} V(M)} \prod_{i=1}^n \text{Tr} M^{k_i}$$

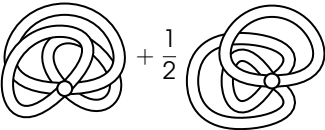
Why this looks like a string theory?

Connected correlators admit a **genus expansion**:

$$\frac{1}{N^n} \left\langle \prod_{i=1}^n \text{Tr } M^{k_i} \right\rangle_{\text{conn}} \sim \sum_{g \geq 0} N^{2-2g-n} \left\langle \prod_{i=1}^n \text{Tr } M^{k_i} \right\rangle_{g, \text{conn}}$$

- Feynman diagrams are ribbon graphs
- Ribbon graphs correspond to 2d surfaces
- $1/N$ counts topology (Euler characteristic)

Example: $V(M) = \frac{M^2}{2}$ (**GUE**)

$$\left\langle \frac{1}{6} : \text{Tr } M^6 : \right\rangle_{g=1, \text{conn}}^{\text{GUE}} = \frac{1}{6} \text{ (diagram 1)} + \frac{1}{2} \text{ (diagram 2)} = \frac{2}{3}$$


Pruned correlators $\Leftrightarrow$ no petals in the Feynman diagram

(’t Hooft, Brezin, Itzykson, Parisi, Zuber)

One-cut matrix models

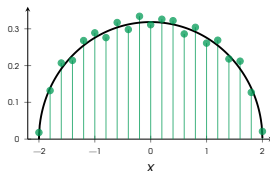
- **Large- N** eigenvalue density $\rho(x)$ supported on a single interval
- Two distinguished points: the edges
- Holds for a large class of potentials V

One-cut assumption

$$\text{supp } \rho = [a, b]$$

Example: $V(M) = \frac{M^2}{2}$ (**GUE**)

$$\rho(x) = \frac{1}{2\pi} \sqrt{4 - x^2}$$



The spectral curve

- In the large- N limit, the resolvent $W(x)$ satisfies

$$\mathcal{S}: E(x, y) = 0, \quad y(x) = V'(x) - 2W(x).$$

This defines the **spectral curve**.

- On the cut:

$$y(x \pm i0) = \mp 2\pi i \rho(x).$$

- For one-cut models, it is rational ($\mathcal{S} \cong \mathbb{CP}^1$) and can be parametrised as

$$\begin{cases} x(z) = \gamma(z + \frac{1}{z}) + \delta \\ y(z) = \frac{\gamma}{2}(z - \frac{1}{z})Q(z) \end{cases}$$

for $\gamma = \frac{b-a}{4}$, $\delta = \frac{a+b}{2}$; Q captures the dependence on V .

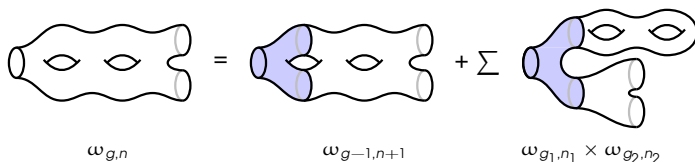
Can the **geometry** of the spectral curve
solve the model?

Topological recursion: geometry $\Rightarrow$ full solution

The correlation differentials

$$\omega_{g,n}(z_1, \dots, z_n) = \left\langle \prod_{i=1}^n \text{Tr} \frac{dx(z_i)}{x(z_i) - M} \right\rangle_{g, \text{conn}}$$

are determined recursively from the spectral curve.



- The recursion is **topological**
- It is **local**: only residues at critical points of x enter

(Chekhov–Eynard–Orantín)

Uniqueness:

Once the spectral curve is fixed,
topological recursion uniquely determines the full
 $1/N$ expansion.

Landau–Ginzburg without gravity

B-twisted Landau–Ginzburg model:

- 1d target space X
- Superpotential $W: X \rightarrow \mathbb{C}$
- Holomorphic top form Ω

Properties:

- Localises on **constant maps** at the critical points of W
- **State space**: chiral ring

$$\mathcal{H} = \text{Jac}(W) = \bigoplus_{dW(\alpha)=0} \mathbb{C}_{\alpha}$$

- Defines a **2d topological field theory**

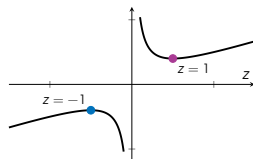
$$\langle \cdot \rangle_g : \mathcal{H}^{\otimes n} \rightarrow \mathbb{C}$$

- Correlators are computed by residue formulas

Example: B-twisted $\mathbb{C}P^1$

$$X = \mathbb{C}^{\times}, \quad W(z) = z + \frac{1}{z}, \quad \Omega = \frac{dz}{z}$$

$$\mathcal{H} = \frac{\mathbb{C}[z]}{(z^2 - 1)} \cong \mathbb{C}_+ \oplus \mathbb{C}_-$$



Critical points of W
label the TFT states

Coupling LG to gravity

$$\text{TFT} \longrightarrow \text{CohFT}$$

What changes:

- Worldsheet metric becomes dynamical
- Integration over $\overline{\mathcal{M}}_{g,n}$

What stays:

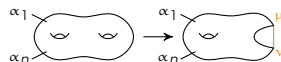
- Localisation on critical points of W
- Same state space

Mathematical output:

- Assignment of cohomology classes

$$\langle\langle \cdot \rangle\rangle_g : \mathcal{H}^{\otimes n} \rightarrow H^*(\overline{\mathcal{M}}_{g,n})$$

- Gluing axioms encode worldsheet factorisation



Degeneration of worldsheets

(Verlinde–Verlinde, Witten, Vafa, Losev, Kontsevich–Manin, Dubrovin,
Dunin-Barkowski–Norbury–Orantin–Popolitov–Shadrin)

What a CohFT means physically

Physical picture:

- CohFT = LG matter sector integrated out
- Worldsheet metric is dynamical
- Integration over $\overline{\mathcal{M}}_{g,n}$

Operators: chiral primaries + gravitational descendants

$$\tau_{\alpha,k} = \mathcal{O}_{\alpha} \times \psi^k, \quad dW(\alpha) = 0, \quad k \geq 0.$$

Correlators:

$$\langle \tau_{\alpha_1,k_1} \cdots \tau_{\alpha_n,k_n} \rangle_g = \int_{\overline{\mathcal{M}}_{g,n}} \langle\langle \mathcal{O}_{\alpha_1} \cdots \mathcal{O}_{\alpha_n} \rangle\rangle_g \prod_{i=1}^n \psi_i^{k_i}$$

CohFTs as a worldsheet Feynman calculus

$$\langle\langle \mathcal{O}_{\alpha_1} \cdots \mathcal{O}_{\alpha_n} \rangle\rangle_g = \sum \text{Diagram}$$

Worksheet Feynman rules:

- **Graphs** $\leftrightarrow$ worldsheets built by gluing
- **Colour** $\leftrightarrow$ localisation at critical point
- **Vertices** $\leftrightarrow$ background
- **Edges** $\leftrightarrow$ gluing rules (nodes)
- **Legs** $\leftrightarrow$ operator insertions (marked points)

Vertices, edges and leg terms defined from **oscillatory integrals**, e.g.

$$\int_{\Gamma_\alpha} e^{\frac{1}{u} W(z)} \Omega$$

(Givental, Pandharipande–Pixton–Zvonkine, Cecotti–Vafa, Nekrasov)

Gauge/string duality

Duality statement

Every one-cut Hermitian matrix model is dual to a B-twisted Landau–Ginzburg theory coupled to gravity

Input dictionary:

- Target space: $X = \text{spectral curve} - \{\text{poles of } x \text{ and } dy\}$
- Superpotential: $W = x = \gamma(z + \frac{1}{z}) + \delta \implies \mathcal{H} = \textcolor{violet}{C}_+ \oplus \textcolor{blue}{C}_-$
- Holomorphic form: $\Omega = dy$

Operator dictionary:

$$\frac{1}{k} : \text{Tr } M^k : \longleftrightarrow \underbrace{\mathcal{O}_+ + (-1)^k \mathcal{O}_-}_{\text{matter insertions}} \times \underbrace{\sum_{m=0}^k \psi^m \prod_{\mu=1}^m \frac{k^2 - (\mu - \frac{1}{2})^2}{2\mu}}_{\text{gravitational descendants}}$$

Example: GUE

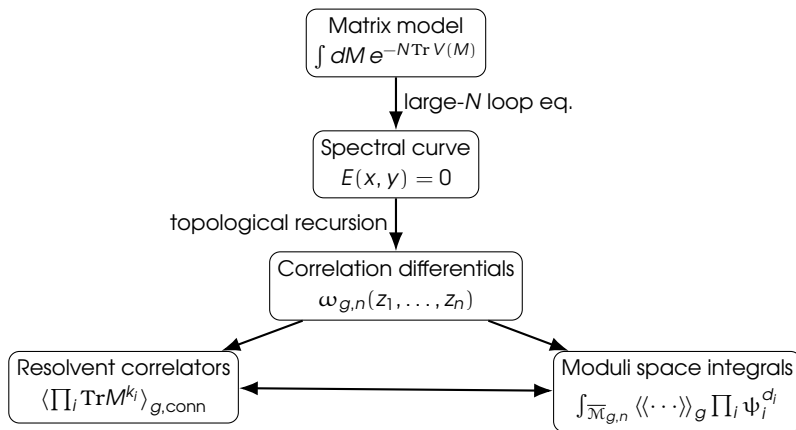
LG model:

$$X = \mathbb{C}^\times, \quad W(z) = z + \frac{1}{z}, \quad \Omega = \frac{1}{2} \left(1 + \frac{1}{z^2} \right) dz, \quad \mathcal{H} = \textcolor{violet}{\mathbb{C}}_+ \oplus \textcolor{blue}{\mathbb{C}}_-$$

Check in $(g=1, n=1)$:

$$\begin{aligned} \left\langle \frac{1}{k} : \text{Tr} M^k : \right\rangle_{g=1, \text{conn}}^{\text{GUE}} &\stackrel{?}{=} \frac{1}{2} \int_{\overline{\mathcal{M}}_{1,1}} \left(\begin{array}{c} 1 + \frac{1}{2} (k^2 - \frac{1}{4}) \psi_1 \\ \text{---} \textcolor{violet}{\bigcirc} \text{---} \\ 1 - \frac{3}{8} \kappa_1 \end{array} + \frac{1}{2} \begin{array}{c} 1 \\ \text{---} \textcolor{violet}{\bigcirc} \text{---} \textcolor{orange}{\bigcirc} \text{---} \\ 1 \quad -\frac{1}{8} \end{array} \right. \\ &\quad \left. + \begin{array}{c} (-1)^k (1 + \frac{1}{2} (k^2 - \frac{1}{4}) \psi_1) \\ \text{---} \textcolor{blue}{\bigcirc} \text{---} \\ 1 - \frac{3}{8} \kappa_1 \end{array} + \frac{1}{2} \begin{array}{c} (-1)^k \\ \text{---} \textcolor{blue}{\bigcirc} \text{---} \textcolor{orange}{\bigcirc} \text{---} \\ 1 \quad -\frac{1}{8} \end{array} \right) \\ &= \frac{1 + (-1)^k}{2} \frac{k^2 - 4}{48}. \end{aligned}$$

Why it works



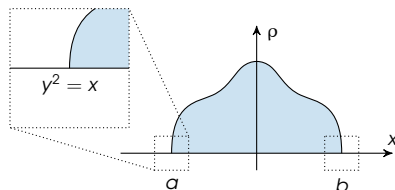
(Eynard, Dunin-Barkowski–Orantin–Shadrin–Spitz)

Berenstein–Maldacena–Nastase limit

Large traces

In the **large trace limit** (BMN-like):

- The spectral curve reduces to the Airy curve
- LG matter decouples from gravity



$$\begin{aligned}
 & \left\langle \prod_{i=1}^n \frac{1}{L_i/t} : \text{Tr } M^{L_i/t} : \right\rangle_{g, \text{conn}} \\
 & \sim t^{-2(3g-3+n)} \times \underbrace{\left(\langle \mathcal{O}_+^n \rangle_g^{\text{TFT}} \pm \langle \mathcal{O}_-^n \rangle_g^{\text{TFT}} \right)}_{\text{pure matter}} \times \underbrace{\int_{\overline{\mathcal{M}}_{g,n}} \prod_{i=1}^n e^{\frac{1}{2} L_i^2 \psi_i}}_{\text{pure top. gravity}}
 \end{aligned}$$

Edge of the spectrum universality

Virasoro and integrability

- Topological recursion $\Leftrightarrow$ Virasoro constraints
- Topological recursion $\rightsquigarrow$ KP integrability

Automatic consequences

- Virasoro constraints on the partition function
- The partition function is a KP tau-function

Integrability is built in

(Dijkgraaf–Verlinde–Verlinde, Kontsevich–Soibelman,
Alexandrov–Bychkov–Dunin-Barkowski–Kazarian–Shadrin)

DSSYK matrix model limits to JT

- ETH matrix model dual to the **DSSYK model**
- One-cut phase with a specific **q -dependent** spectral curve
- Combined limit: large traces and $q \rightarrow 1$

Result (Okuyama conjecture)

For $q = e^{-\lambda}$,

$$\left\langle \prod_{i=1}^n \frac{1}{L_i/\lambda} : \text{Tr } M^{L_i/\lambda} : \right\rangle_{g, \text{conn}}^q \sim \lambda^{-2(3g-3+n)} \times \underbrace{\int_{\overline{\mathcal{M}}_{g,n}} e^{2\pi^2 \kappa_1} \prod_{i=1}^n e^{\frac{1}{2} L_i^2 \psi_i}}_{\text{WP volumes}}$$

From matrices to worldsheets

Every one-cut matrix model is dual to
a Landau–Ginzburg theory coupled to 2d gravity

Main properties:

- Exact **operator dictionary**
- Agreement of correlators at **all genus**
- **No double-scaling** required

Open directions:

- A fully geometric definition of LG gravity (moduli stack, virtual fundamental class)
- Beyond one-cut: higher genus curves
- Extensions to more general gauge theories

Thank you for the attention!